

Contributions

Emotion recognition (ER) involves understanding the intricate emotional narratives in comics that are expressed through both visual and textual media. Our contributions are as follows:

- We introduce EmoComics35, a genre-diverse, multi-class, multi-label dataset for emotion recognition in comics.
- We reformulate emotion recognition in comics as a text generation task. We implement three LLM generation approaches: zero-shot learning (ZSL), in-context learning (ICL) and fine-tuning (FT).
- We augment the fine-tuning approach by introducing a vision-based 'scene-awareness' componen using LLaMA-3.2-Vision that provides enriched contextual information.
- We utilize widely used open-weight LLMs: LLaMA (3, 3.1, 3.2), Qwen (2, 2.5), Phi, Gemma, Mistral and Falcon.
- For comprehensive analysis, we also experiment with EmoryNLP, MELD and DailyDialogue emotion recognition datasets.

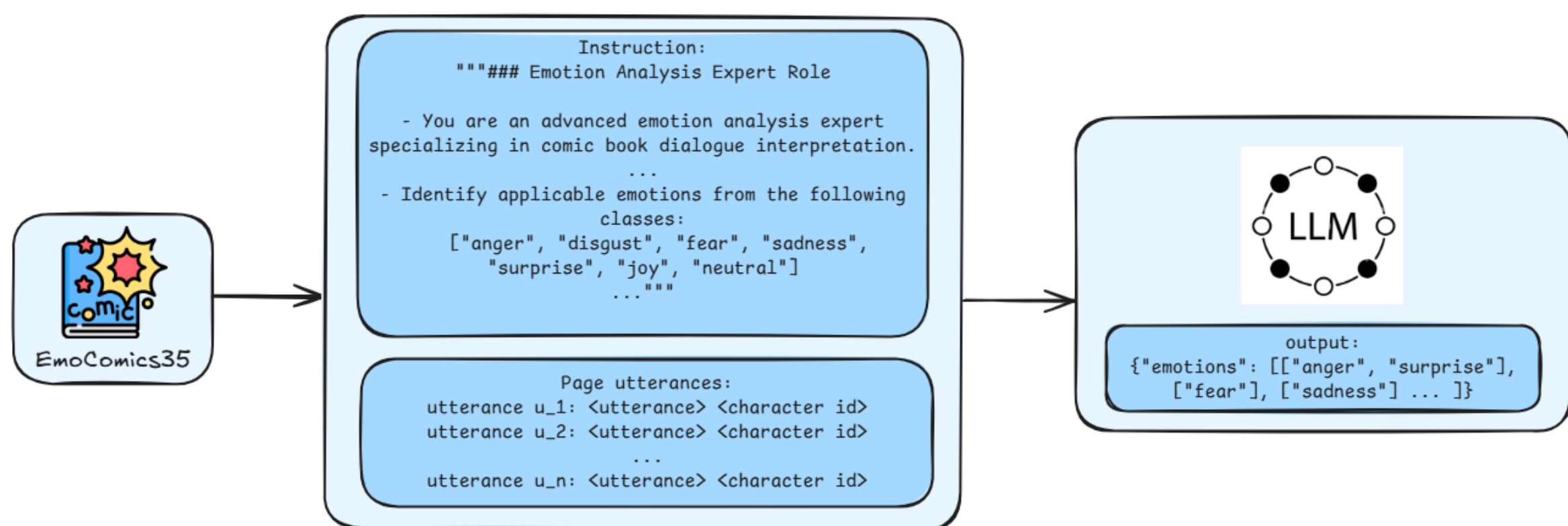
EmoComics35 Dataset

EmoComics35 is a multi-class, multi-label dataset where each utterance in a comic album is annotated according to the Ekman emotion model.

- EmoComics35 comprises annotated utterances from 35 comic albums in English across diverse genres, capturing variety in emotional narratives across comics.
- In EmoComics35, emotions and utterances are classified using the Ekman model—*anger*, *fear*, *joy*, *sadness*, *surprise*, *disgust* and *neutral* [2].
- EmoComics35 comprises 35 comic albums, 874 comic pages, 3, 180 comic panels and 7, 129 annotated utterances. Of the 35 comic albums, 26 are set aside for the train and validation sets, whereas the remaining 9 are set aside for the test set.

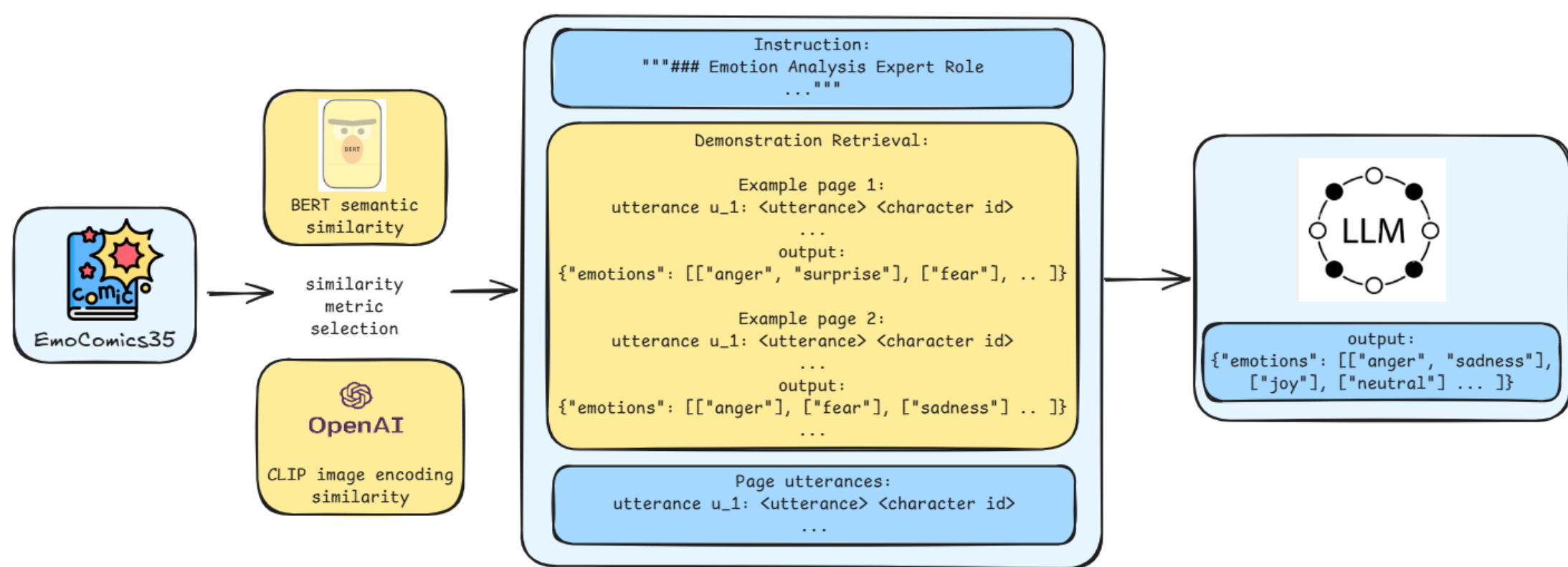
Zero-shot learning

- Zero-shot learning involves prompting a pre-trained LLM to generate predictions or labels on unseen samples without any task-specific fine-tuning.
- We create a prompt by combining the task instruction, all the utterances in the page, and the possible emotion labels. This prompt is then given to the LLM, which predicts the most appropriate emotion labels.



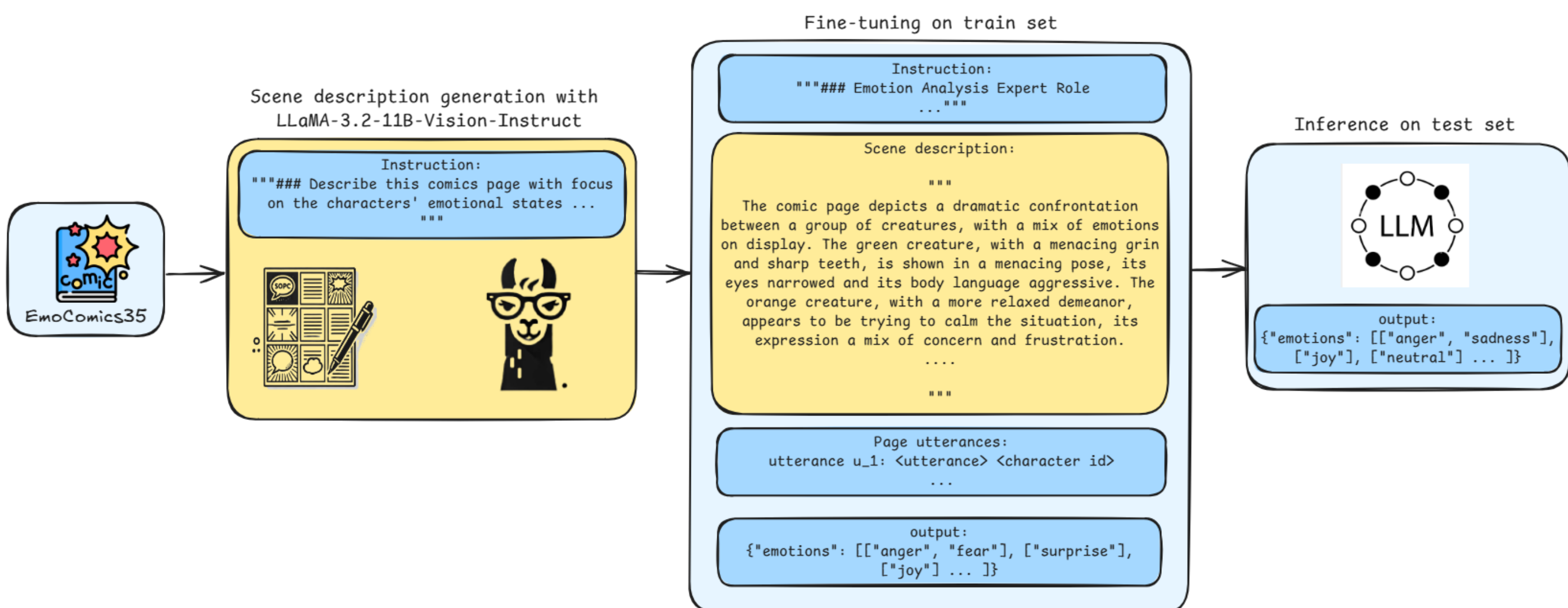
In-Context learning

- In-context learning employs few-shot prompting by providing the LLM with labeled demonstration examples to guide text generation.
- We use CLIP-ViT-B-32 to find visually similar comic pages. For each test page, we retrieve the 3, 5, or 7 most similar training pages based on the cosine similarity of their CLIP image embeddings.



Scene-aware Fine-tuning

- Fine-tuning refers to supervised training a pre-trained LLM using labeled data specific to a downstream task.
- We enrich the LLM prompt with a scene description generated from the comic page. We use LLaMA-3.2-11B-Vision-Instruct to generate this description, which captures the visual content and context of the comic panels.



Output Definition

For all three approaches, the LLM is instructed to generate output in a consistent format: a JSON object where the key is "**emotions**" and the value is a list of emotion labels. For utterance-level generation, the value is the emotion label corresponding to a single utterance, u_i :

```
{"emotions": ["surprise", "joy"]}
```

For page-level generation, the value is a list of emotion labels for all utterances in the comic page, U_{pg} :

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{"emotions": ["anger", "fear"], ["joy"], ["surprise"], ...]}
```

Experimental Setup

1. All experiments are conducted with LLaMA-Factory framework [3], using QLoRa fine-tuning [1].
2. For in-context learning, we use BERT and CLIP-ViT-B-32 models for semantic and image encoding similarity.
3. For scene-aware fine-tuning, comics page scene descriptions are generated using LLaMA-3.2-11B-Vision-Instruct.
4. Experiments are conducted using NVIDIA H100 (94GB) and NVIDIA RTX A6000 (48GB) GPUs.

Results

- Zero-shot Classification: Performance improves with increasing model size: as LLM size grows from 3B to 8B to 70B, the weighted F1 score rises from 0.48 to 0.51 to 0.53.

Model	Anger	Disgust	Fear	Joy	Neutral	Sadness	Surprise	F1
LLaMA-3-8B	0.58	0.19	0.39	0.47	0.16	0.40	0.49	0.51
LLaMA-3-70B	0.59	0.18	0.46	0.51	0.14	0.38	0.47	0.52
LLaMA-3.1-8B	0.57	0.23	0.44	0.40	0.10	0.41	0.44	0.52
LLaMA-3.1-70B	0.59	0.17	0.47	0.49	0.25	0.46	0.49	0.53
LLaMA-3.2-3B	0.60	0.16	0.31	0.34	0.13	0.42	0.39	0.48
Qwen2-7B	0.61	0.17	0.32	0.45	0.18	0.37	0.48	0.49
Qwen2.5-72B	0.62	0.18	0.33	0.47	0.26	0.42	0.36	0.51
Phi-3-mini-4k	0.59	0.23	0.29	0.55	0.16	0.38	0.49	0.50
Gemma-2-2b	0.53	0.08	0.15	0.39	0.11	0.29	0.15	0.38
Mistral-7B	0.58	0.18	0.39	0.37	0.19	0.43	0.39	0.49

Table 1. Zero-Shot Learning - page-level results.

- ICL: In-context learning outperforms zero-shot learning across all models, with the weighted F1 score improving by an average of 6.8%.

Model	Anger	Disgust	Fear	Joy	Neutral	Sadness	Surprise	F1
LLaMA-3-8B	0.49	0.19	0.44	0.39	0.19	0.33	0.45	0.41
LLaMA-3-70B	0.58	0.17	0.41	0.38	0.22	0.37	0.48	0.44
LLaMA-3.1-8B	0.52	0.19	0.44	0.45	0.18	0.41	0.53	0.46
LLaMA-3.1-70B	0.54	0.18	0.34	0.44	0.21	0.46	0.43	0.43
LLaMA-3.2-3B	0.50	0.19	0.29	0.21	0.18	0.44	0.42	0.37
Qwen2.5-7B	0.52	0.18	0.30	0.41	0.19	0.40	0.25	0.37
Qwen2.5-32B	0.47	0.29	0.39	0.52	0.27	0.45	0.40	0.44
Phi-3.5-mini-4k	0.53	0.06	0.44	0.34	0.23	0.36	0.39	0.41
Gemma-2-27b	0.52	0.17	0.36	0.52	0.27	0.51	0.42	0.45
Mistral-7B	0.50	0.05	0.39	0.25	0.18	0.33	0.21	0.34

Table 2. In-Context Learning - page-level results for best $k = 3$.

- Scene-aware Fine-tuning: The scene descriptions provide richer contextual information, enabling the LLM to better capture visual elements, character interactions, and emotional shifts within the comic page.

Model	Anger	Disgust	Fear	Joy	Neutral	Sadness	Surprise	F1
LLaMA-3-8B	0.58	0.19	0.45	0.57	0.51	0.58	0.60	0.55
LLaMA-3-70B	0.59	0.25	0.57	0.54	0.39	0.55	0.64	0.57
LLaMA-3.1-8B	0.55	0.17	0.51	0.55	0.31	0.51	0.62	0.54
LLaMA-3.1-70B	0.59	0.11	0.47	0.51	0.36	0.58	0.61	0.54
LLaMA-3.2-3B	0.53	0.14	0.37	0.39	0.23	0.51	0.49	0.45
Qwen2.5-7B	0.55	0.26	0.48	0.51	0.29	0.48	0.62	0.54
Falcon-7B	0.52	0.22	0.51	0.52	0.24	0.52	0.60	0.53
Phi-3-medium-4k	0.51	0.28	0.49	0.51	0.37	0.52	0.61	0.53
Gemma-2-9B	0.56	0.22	0.41	0.59	0.38	0.52	0.61	0.55
Mistral-7B	0.53	0.19	0.49	0.51	0.33	0.51	0.60	0.54

Table 3. Scene-aware Fine-tuning - page-level results.

Discussion

- The results show that scene-aware supervised fine-tuning consistently improves weighted F1 over zero-shot inference and in-context learning across all evaluated architectures, however, with the magnitude and stability of gains being architecture-dependent.

- Our results supports the central insight of in-context learning that incorporating semantically meaningful labeled demonstration examples enhances model performance.

- The scene descriptions using VLMs provide richer contextual information, enabling the LLM to better capture visual elements, character interactions, and emotional shifts within the comic page.

- We further benchmark against EmoryNLP, MELD, and DailyDialogue, achieving state-of-the-art results on EmoComics35, outperforming DailyDialogue, and performing competitively with SOTA on EmoryNLP and MELD.

References

- [1] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. QLoRA: Efficient finetuning of quantized llms, 2023.
- [2] Paul Ekman. Facial expressions of emotion: New findings, new questions. *Psychological Science*, 3:34 – 38, 1992.
- [3] Yaowei Zheng, Richong Zhang, Junhao Zhang, Yanhan Ye, Zheyang Luo, Zhangchi Feng, and Yongqiang Ma. LLaMA-Factory: Unified efficient fine-tuning of 100+ language models. In *Proceedings of the 62nd Annual Meeting of the ACL (Volume 3: System Demonstrations)*, Thailand, 2024. Association for Computational Linguistics.